

```
In [ ]: import pandas as pd
import numpy as np
import joblib

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

```
In [ ]: df = pd.read_csv("customer_booking.csv", encoding="latin-1")

df.head()
```

```
Out[ ]: 
```

	num_passengers	sales_channel	trip_type	purchase_lead	length_of_stay	flight_hour
0	2	Internet	RoundTrip	262	19	7
1	1	Internet	RoundTrip	112	20	3
2	2	Internet	RoundTrip	243	22	17
3	1	Internet	RoundTrip	96	31	4
4	2	Internet	RoundTrip	68	22	15

```
In [ ]: # encoding
categorical_cols = [
    'sales_channel',
    'trip_type',
    'flight_day',
    'route',
    'booking_origin'
]

encoders = {}

for col in categorical_cols:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])

    encoders[col] = le
```

```
In [ ]: # features and target
X = df.drop("booking_complete", axis=1)
y = df["booking_complete"]
```

```
In [ ]: # train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

```
In [ ]: from sklearn.ensemble import RandomForestClassifier
```

```
rf_model = RandomForestClassifier(  
    n_estimators=100,      # trees kam  
    max_depth=10,         # depth limit  
    min_samples_split=5,  
    min_samples_leaf=2,  
    random_state=42,  
    n_jobs=-1  
)  
  
rf_model.fit(X_train, y_train)
```

```
Out[ ]: ▼ RandomForestClassifier ⓘ ?  
        ► Parameters
```

```
In [ ]: # accuracy  
y_pred = rf_model.predict(X_test)  
  
accuracy = accuracy_score(y_test, y_pred)  
  
print("Accuracy:", accuracy)
```

Accuracy: 0.8506

```
In [ ]: import joblib  
  
joblib.dump(  
    rf_model,  
    "british_airways_rf_model.pkl",  
    compress=9  
)  
  
print("Model Saved")
```

Model Saved

```
In [ ]: import os  
  
size_mb = os.path.getsize("british_airways_rf_model.pkl") / (1024 * 1024)  
  
print(f"Model Size: {size_mb:.2f} MB")
```

Model Size: 1.97 MB

```
In [ ]: joblib.dump(  
    encoders,  
    "label_encoders.pkl"  
)  
  
print("Encoders Saved")
```

Encoders Saved

```
In [ ]:
```